Midterm Report

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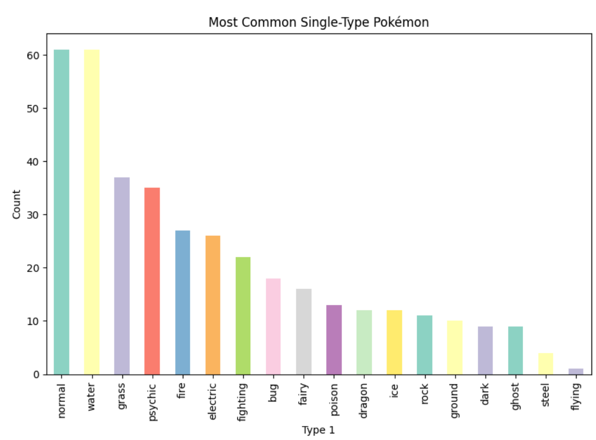
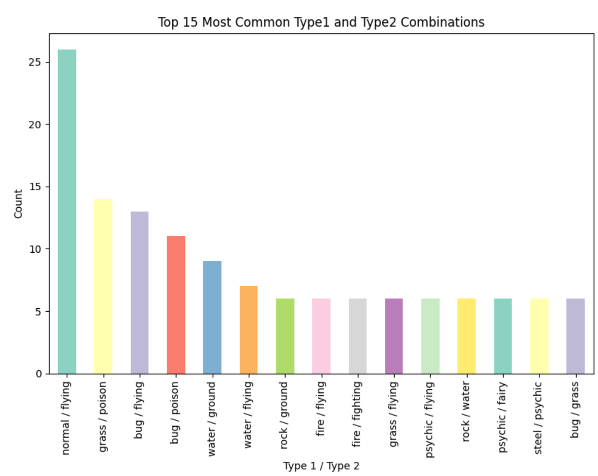
Data Exploration

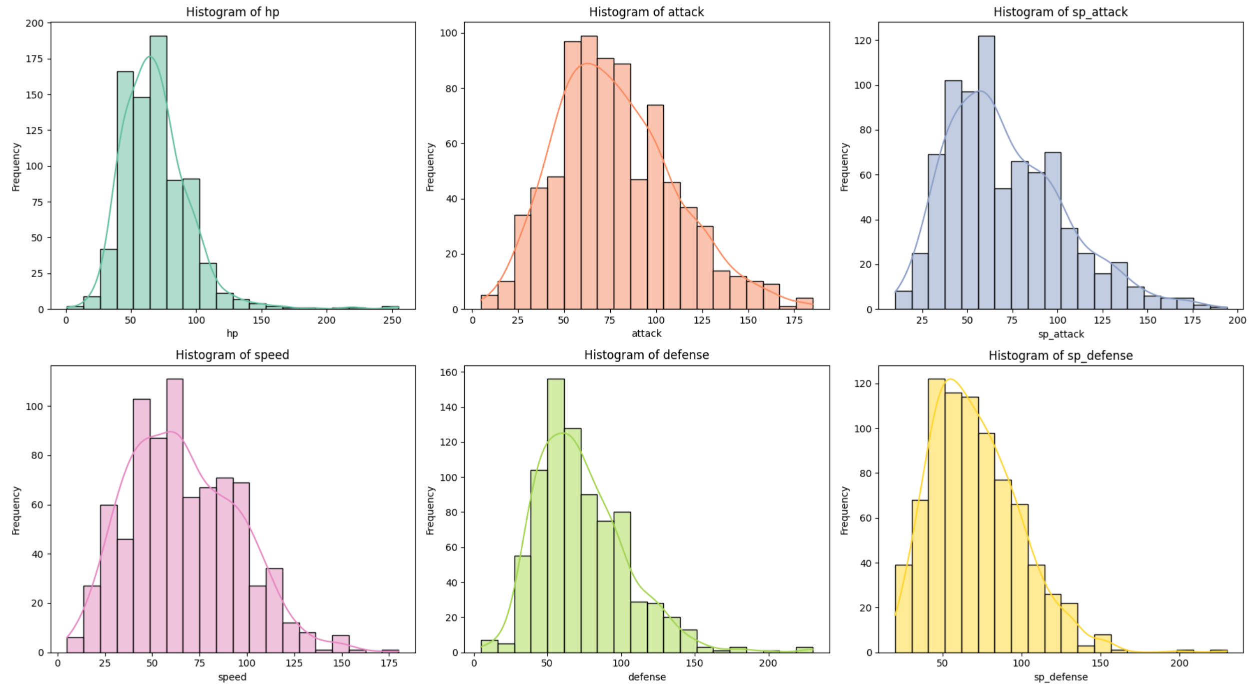
We used the complete Pokemon Dataset, authored by Rounak Banik who compiled the information from the well known online Pokemon encyclopedia Serebii. The dataset contains all 802 pokemon from generation 1-7 (Kanto-Alola), and includes data on the Pokemon's name, base stats, types matchup, height, weight, shape classification, egg steps, experience points, abilities, stats, etc. This information itself comes from the national pokedex in the games themselves.

The first analysis we did was to check what were the most common Pokemon single and double typings. For single typings, we found that the top 5 were Normal, Water, Grass, and Psychic, while for dual typings they were Normal/Fly, Grass/Poison, Bug/Fly, Bug/Poison, and Water/Ground. Interesting to note is that while there is nearly no solo flying type pokemon, there is a massive amount of dual type pokemon.

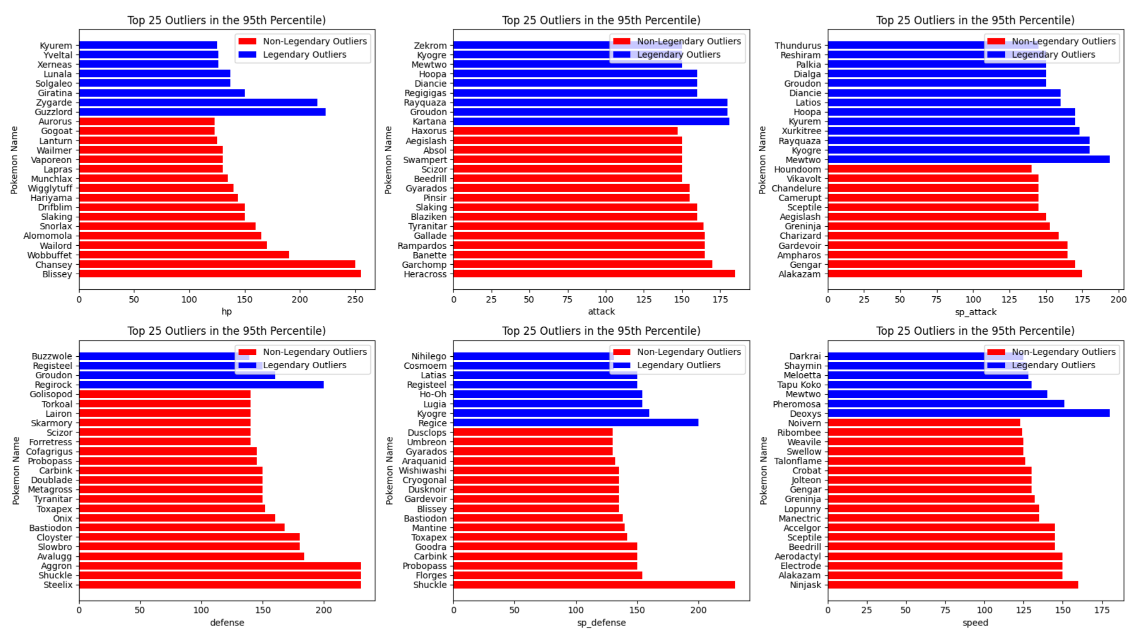




Next, we wanted to begin to explore what the average pokemon's stats look like. For those that play Pokemon, note that we are only looking at base stats in this analysis and will be ignoring Evs and Ivs, which allow for some wild stat maximizing when utilized correctly. Here are the results we found examining for the averages:

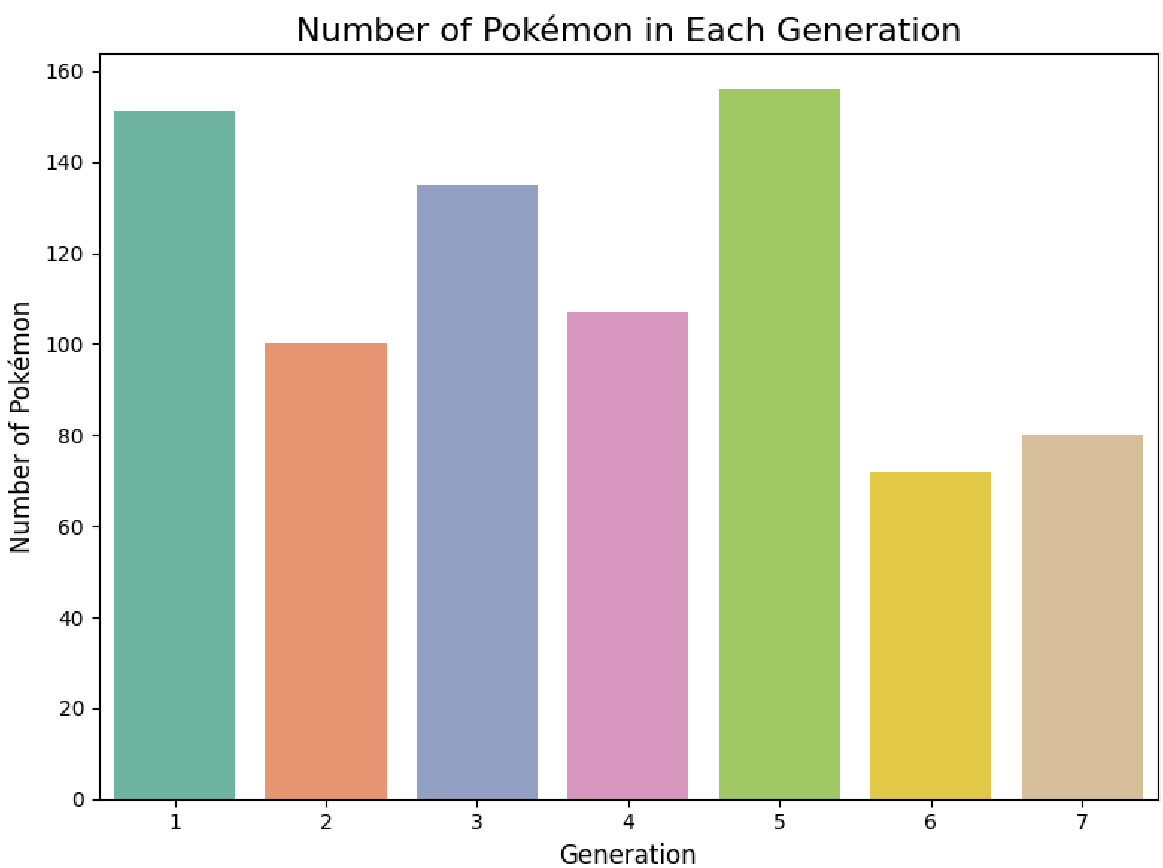


In general, the averages are all around 60. It is interesting to note however that both attack and special attack have more pokemon with high tier stats compared to the others. Zooming in on that section, we then examined what pokemon actually make up the upper 95th percent of each stat:



As we can see in blue, the median number of legendaries that show up in the top 25 pokemon of the 95th percentile for each stat category is 8, corresponding to nearly a third of the top 25 despite only making up 8.7% of all total population. The other interesting tidbit is for two thirds of the battle stats, non-legendaries take the top spot as the pokemon with the highest base stat, with Blissey, Heracross, Steelix, and Shuckle winning out for hp, attack, defense and special defence. In contrast, Mewtwo and Deoxys take the top spots for special attack and speed respectively.

Next, we wanted to explore how the Pokemon Co. populated each generation with Pokemon and what their distributions were. As we can see below, gen 1 and gen 5 both have the most pokemon added with 151 each. This makes sense as Unova was meant to pay homage to Kanto with its whole new slate of pokemon. Interestingly, within the first 5 generations, gen 3 has the third most newly added pokemon. Again, this makes Johto (gen 2) primarily made of left over designs from gen 1, and gen 3 marked a shift from the original gameboy to the gameboy advanced, freeing up more space and potential for new pokemon. Gen 6 has the least amount of new pokemon, but this also makes sense since it was the first game to be made in full 3d and they did not have any assets they could reuse.



We also wanted to see how the demographics of what types of pokemon were added changed over time. In general water, normal, grass, and bug almost always make up half of the pokemon types, as they are common and are prevalent in the early game. Specific shout outs go to generation 2 for adding steel and dark, and generation 6 for introducing fairy. In general ice, ghost, and steel types are the most rare generation to generation.